Momentum in two dimensions

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Conserve momentum separately along perpendicular axes.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 7.*
East positive: 5-3=2 kg·m/s east. Opposite directions require opposite signs.

02. Worked example: explain the model
Right triangle with legs 3 and 4; resultant 5 northeast.

03. Read and interpret the model
Result points northeast; ordinary lengths are not diagonal magnitude.

04. Apply the physics
3 m/s east and 4 m/s north.

05. Explain your reasoning
Otherwise the two equations may describe different interactions or omit parts of the system; conservation comparison becomes inconsistent.

06. Review: connect an earlier idea *Review: Unit 2, Day 14.*
Fastest at 4 AU. Sweeping equal areas in equal times requires greater speed near the Sun.

07. Exit check
2 m/s west and 1 m/s south.

08. STRETCH: test the model
Components 3 m/s east and 4 m/s north; supplied 3-4-5 relationship gives speed 5 m/s.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 7.*
Deviation≈3.333%. It compares values relative to the initial value; uncertainty requires a measurement/error analysis.

02. Worked example: explain the model
Right triangle with legs 3 and 4; resultant 5 northeast.

03. Read and interpret the model
Result points northeast; ordinary lengths are not diagonal magnitude.

04. Apply the physics
Components (2, 4/3) m/s; speed 2.40 m/s; angle 33.7°.

05. Explain your reasoning
Otherwise the two equations may describe different interactions or omit parts of the system; conservation comparison becomes inconsistent.

06. Review: connect an earlier idea *Review: Unit 2, Day 14.*
$T_B/T_A=4^{(3/2)}\approx 8$. The normalization depends on the central mass and chosen units.

07. Exit check
(6,-3) kg·m/s;√45=6.71 kg·m/s.

08. STRETCH: test the model
Total momentum = (4, 6) kg·m/s; total mass = 4 kg. v_f = (1, 1.5) m/s; speed = √3.25 ≈ 1.80 m/s; angle = arctan(1.5) ≈ 56.3° north of east.